



DATA SCIENCE FIELD NOTES / 06B

WEEK 6 PROJECT SUMMARY

From problem definition to reproducible preparation and initial no-show prediction

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	HealthConnect No-Show Prediction
MILESTONE	13 September 2026 Week 6 Batch D

Educational fictional-clinic project. Not a clinical decision system or validated operational tool.

Week 6 project summary

I planned to improve the Week 5 baseline, investigate its errors and integrate a candidate with a reproducible scoring component. I completed temporal validation of four specifications, a no-history feature refinement, error analysis, a workload-threshold scenario and a serialized pipeline with an executable raw-input interface.

Result	Evidence
Candidate	No-history logistic regression
Internal validation AUC	0.6481 mean across three temporal folds
Week 5 baseline internal AUC	0.6416 on the same folds
Reused comparison at 0.5	AUC 0.6653; recall 67.18%; precision 58.56%
Errors at 0.5	184 false positives; 127 false negatives
Scenario threshold 0.600	193 flags; precision 70.47%; recall 35.14%
Integration	17 executable tests passed

What changed from Week 5

I removed history-derived predictors because their availability was uncertain and the simpler candidate ranked highest internally. I retained the original source and Week 5 benchmark. On the reused comparison, the candidate catches more no-shows but generates more false positives at 0.5, while its AUC is slightly lower. The evidence supports further testing, not a claim of universal predictive improvement.

Integration contribution

At my request, ChatGPT simulated the ML Engineering track. I received input/output and validation requirements and provided the candidate and model requirements. Applying that review produced a callable scoring interface, a bundled preprocessing/model artifact, warnings and rejection checks. This is AI-simulated integration evidence; no human intern exchange is claimed.

Remaining issues and next stage

The main challenges were feature timing uncertainty, a previously examined test set, inconsistent patient histories, Sunday scheduling and future cancellations. Removing history mitigates one timing risk; it does not repair the underlying source data. The workload target is hypothetical and requires stakeholder agreement.

Important decisions

I selected the candidate from internal temporal validation rather than the old test set. I did not select the random forest despite its higher reused-test accuracy, because it ranked lower on the predeclared internal AUC criterion. I serialized a scenario threshold separately from the model so that its meaning and future revision are explicit.

Week 7 priorities

Obtain fresh evaluation data; validate the interface on realistic batches; establish capacity and follow-up procedures; assess calibration and subgroup errors with adequate support; revisit cancellation handling; and measure technical reliability before a pilot. The candidate should remain frozen while these tests are designed.

Submission and collaboration status

The technical package, summary and AI-simulated integration evidence are complete. The programme must confirm whether the simulation is accepted for its mandatory cross-track requirement. I have not claimed a human collaboration, deployed the system, uploaded to Google Drive, committed to GitHub or submitted the official form.

Contribution to HealthConnect

Week 6 turns a notebook baseline into a documented, locally integrated component. The useful progress is clearer validation, fewer unverified inputs, explicit operating trade-offs and executable evidence for the next testing stage.